Linear Regularisation, Latent Projections, and Neural Ensembles for Predicting Aqueous Solubility from Molecular Descriptors

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***Abstract*—Predicting aqueous solubility directly from molecular structure is a central regression problem in computational chemistry and early-stage drug discovery. This study evaluates four regression models on a dataset of $1{,}267$ chemical compounds (951 training and 316 test observations) characterised by 228 collinear molecular descriptors subjected to Yeo-Johnson transformation and standardisation. We implement ordinary least squares (OLS), L_2-penalised ridge regression, and partial least squares (PLS) via the NIPALS algorithm from scratch, comparing them against a regularised multilayer perceptron (MLP) ensemble using cross-validation and paired bootstrap resampling. While ridge regression ($\mathrm{RMSE} = 0.7175$, $R^2 = 0.8805$), PLS ($\mathrm{RMSE} = 0.7337$, $R^2 = 0.8750$), and OLS ($\mathrm{RMSE} = 0.7430$, $R^2 = 0.8718$) perform comparably on the test set without statistically significant differences among them, the nonlinear neural network ensemble achieves a test RMSE of 0.6407 ($R^2 = 0.9047$), outperforming all linear estimators even after Bonferroni correction.**

***Index Terms*—aqueous solubility, ordinary least squares, ridge regression, partial least squares, neural networks**

I. Introduction

Aqueous solubility governs compound bioavailability, membrane permeation, and formulation viability in pharmaceutical development [1]. Because experimental assays across large chemical libraries are costly and time-consuming, quantitative structure-property relationship (QSPR) models seek to estimate logarithmic solubility directly from numerical molecular descriptors. In statistical learning terms, this constitutes a supervised multivariate regression problem in which a continuous response $y \in \mathbb{R}$ is approximated from a predictor vector $\mathbf{x} \in \mathbb{R}^D$.

Chemical descriptor matrices present two classical challenges for regression estimators: severe skewness in topological counts and near-exact multicollinearity among structural fingerprints [2]. Unregularised ordinary least squares (OLS) yields unbiased linear predictions under standard Gauss-Markov assumptions, but its coefficient variance inflates rapidly as singular values of the design matrix approach zero [3]. Two distinct linear strategies address this ill-conditioning: continuous shrinkage via L_2 regularisation (ridge regression) [4] and supervised dimension reduction onto orthogonal latent factors via partial least squares (PLS). However, linear projections cannot capture higher-order interactions among functional groups unless nonlinear mappings, such as feedforward neural networks, are introduced.

In this work, we implement OLS, ridge regression, and PLS from scratch, verify their numerical agreement against reference implementations [5], and compare their generalisation capacity against a regularised multilayer perceptron (MLP) ensemble. Section II details the dataset pre-processing, mathematical formulations, and resampling protocols; Section III presents the cross-validation profiles, test-set predictions, and paired bootstrap comparisons; and Section IV summarises our findings.

II. Methods

A. Dataset and pre-processing

The data are the solubility set distributed with [1], with $N = 1{,}267$ chemical compounds split into 951 training and 316 test observations. Each compound is described by $D = 228$ predictors: 208 binary fingerprints, 16 count descriptors and 4 continuous descriptors. The outcome is the log solubility of the compound.

All 20 non-binary predictors are right-skewed, so they receive a Yeo-Johnson transformation [6]; the binary predictors are left unchanged. Every predictor is then centred and scaled to unit standard deviation. The transformation parameters, means and standard deviations are estimated on the fitting data only and re-applied, without re-estimation, to the data being evaluated. The transformed predictors are strongly collinear: 47 pairs have $|r| > 0.9$, and `FP134` and `NumSulfer` are nearly duplicates ($r = 1.000$ to three decimals).

B. Ordinary least squares

With $\mathbf{X}$ centred on the fitting data and $\mathbf{y}_c = \mathbf{y} - \bar{y}$, the intercept equals $\bar{y}$ and the ordinary least squares (OLS) coefficients are

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top \mathbf{y}_c = \mathbf{X}^+ \mathbf{y}_c, \tag{1}$$

where $(\cdot)^+$ is the Moore-Penrose pseudoinverse [3]. The estimator was implemented from scratch through the singular value decomposition $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, discarding singular values below $\max(n, p)\, \varepsilon\, s_{\max}$. Working on $\mathbf{X}$ avoids squaring its condition number, and the pseudoinverse returns the minimum-norm solution when the rank drops. OLS has no tuning parameter and is directly interpretable, but its coefficients become unstable under collinearity [2].

C. Ridge regression

Ridge regression [4] penalises the squared norm of the coefficients and leaves the intercept unpenalised. With the same centring, the solution is

$$\hat{\boldsymbol{\beta}}(\lambda) = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}_c = \mathbf{V}\, \mathrm{diag}\left(\frac{s_i}{s_i^2 + \lambda} \right) \mathbf{U}^\top \mathbf{y}_c. \tag{2}$$

The form on the right, also implemented from scratch, needs a single decomposition for the whole search. The penalty shrinks the directions of small singular value and so stabilises the coefficients, at the cost of a bias and of having to select λ. We search 29 values spaced logarithmically between 10^{-3} and 10^4 and take the value that minimises the mean cross-validated RMSE.

D. Resampling and performance measures

Cross-validation uses the training set only, in 10 folds of 95 or 96 compounds drawn with a fixed seed and shared by the OLS, ridge, and PLS analyses. In each fold, the pre-processing and the model are estimated on the other nine folds and applied to the held-out fold. Performance is measured by RMSE and by $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$, averaged over the folds, with the standard error of the mean. Both the models and the cross-validation were compared with `LinearRegression`, `Ridge`, `cross_validate` and `GridSearchCV` from scikit-learn [5], with the pre-processing placed inside a `Pipeline` to preserve the separation between folds. Test-set uncertainty is expressed by 95% percentile bootstrap intervals [7] over 5000 resamples of the 316 test compounds.

E. Partial least squares and neural network

Partial least squares (PLS) constructs $A \le D$ orthogonal latent scores $\mathbf{t}_a = \mathbf{X}_{a-1}\mathbf{w}_a$ that maximise the sample covariance with the deflated response [2]. We implemented the univariate NIPALS algorithm from scratch: at step a, weights $\mathbf{w}_a = \mathbf{X}_{a-1}^\top \mathbf{y}_{a-1} / \|\mathbf{X}_{a-1}^\top \mathbf{y}_{a-1}\|_2$ produce scores $\mathbf{t}_a$, loadings $\mathbf{p}_a = \mathbf{X}_{a-1}^\top \mathbf{t}_a / (\mathbf{t}_a^\top \mathbf{t}_a)$, and response weights $q_a = \mathbf{y}_{a-1}^\top \mathbf{t}_a / (\mathbf{t}_a^\top \mathbf{t}_a)$, followed by rank-one deflation of $\mathbf{X}_{a-1}$ and $\mathbf{y}_{a-1}$. Accumulating A components yields the direct regression vector $\hat{\boldsymbol{\beta}}_{\mathrm{PLS}}(A) = \mathbf{W}(\mathbf{P}^\top \mathbf{W})^{-1}\mathbf{q}$. We evaluate $A \in \{1, \ldots, 40\}$ via 10-fold CV. PLS handles collinearity via supervised projection and reduces dimensionality, though latent factors mix all original descriptors.

To capture nonlinear interactions, we train feedforward multilayer perceptrons (MLPs) with ReLU activations, dropout, and L_2 weight decay using AdamW ($\eta = 10^{-3}$, batch size 64) on standardised targets. Because neural networks are sensitive to hyperparameter choices and random weight initialisation, we perform a 5-fold cross-validation grid search over architectures $\{(16,), (64,), (128,), (64, 32)\}$, weight decay $\{10^{-2}, 10^{-1}, 1.0\}$, dropout $\{0.0, 0.3\}$, and training epochs from 25 to 300 in steps of 25. The final predictor averages an ensemble of 5 independently initialised networks trained on the full training set under the selected configuration, reducing variance at the cost of higher computational effort and lower interpretability.

III. Results and Discussion

A. Ordinary least squares

Table I summarises the OLS results. On the full training set $\mathbf{X}$ has numerical rank 228 of 228 and a condition number of about 3.1×10^4: it is badly conditioned but not exactly singular. The collinearity is visible in the coefficients. `NumSulfer` receives -106.14 and `FP134` receives $+105.95$, two values that nearly cancel, so that only their sum is well determined by the data.

TABLE I
OLS performance. CV values are the mean over 10 folds ± the standard error; the test interval is a 95% bootstrap interval.

Set	RMSE	R^2
Training	0.4900	0.9426
10-fold CV	0.7279 ± 0.0215	0.8654 ± 0.0128
Test	0.7430	0.8718
Test interval	[0.6645, 0.8243]	[0.8327, 0.9027]

The implementation agrees with scikit-learn to numerical precision. On the test set both give an RMSE of 0.742978 and an R^2 of 0.871814, with a largest difference of 1.5×10^{-10} in the coefficients and of 5.4×10^{-13} in the predictions. The two cross-validations differ by at most 1.3×10^{-11} in the RMSE of any fold. In folds 4 and 7 the rank falls to 227, since removing some compounds makes one column an exact combination of the others; this is where the pseudoinverse of (1) is indispensable.

The gap between the training RMSE of 0.4900 and the test RMSE of 0.7430 indicates overfitting, with 228 coefficients estimated from 951 compounds. Cross-validation anticipated the test performance well, as Fig. 1 shows. The test RMSE lies 0.0151 above the CV mean, less than one standard error, and inside the 95% interval of the CV mean, [0.679, 0.777]. The test R^2 likewise lies inside the corresponding interval, [0.837, 0.894]. The RMSE of individual folds ranges from 0.611 to 0.853, and the test points follow the diagonal across the whole solubility range.

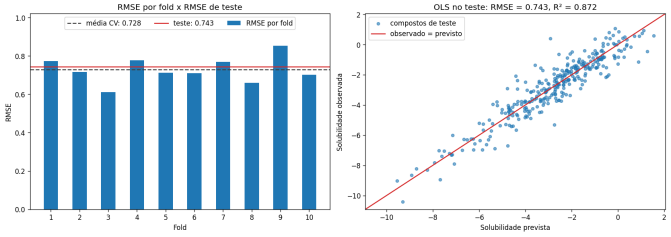


Fig. 1. OLS: RMSE per fold against the CV mean and the test RMSE (left), and observed against predicted solubility on the test set (right).

B. Ridge regression

The solution of (2) matches the normal equations written with an explicit unpenalised intercept to within 7.1×10^{-14}, and the intercept obtained is exactly $\bar{y}$. Across the 29 values of λ it also matches scikit-learn, with largest differences of 7.4×10^{-9} in the coefficients, at $\lambda = 10^{-3}$, and of 1.2×10^{-11} in the training predictions.

Fig. 2 shows the cross-validation profile. For small λ the RMSE is that of OLS (0.7278). It rises slightly to about 0.735 near $\lambda = 0.1$, falls to its minimum at $\lambda \approx 31.6$, and then grows quickly to 1.3865 at $\lambda = 10^4$, where the model is over-penalised. The minimum lies inside the grid, so the search range was sufficient. GridSearchCV selects the same λ, with a largest difference in the profiles of 3.4×10^{-4} in the RMSE. This difference has an identified cause: `StandardScaler` divides by n and our pre-processing by $n - 1$. Rescaling the columns does not change the OLS predictions, but in ridge it is equivalent to a slight change of λ. Under the same convention the difference falls to 1.1×10^{-13}.

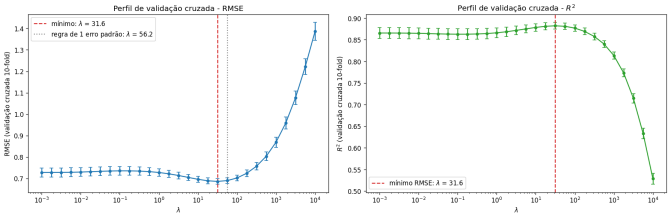


Fig. 2. Ridge cross-validation profile of RMSE (left) and R^2 (right), with the standard error of the mean over 10 folds.

Table II reports the final model, refitted on the whole training set. Scikit-learn gives the same test result, with a largest difference of 4.6×10^{-14} in the predictions. Relative to OLS, the penalty reduced the test RMSE by 0.0255 and the norm of the coefficients from 152.02 to 1.48. The collinear pair is the clearest case: `NumSulfer` and `FP134` become -0.057 and -0.056, sharing the weight between two nearly duplicate predictors instead of opposing it.

The cross-validated estimate is more optimistic than for OLS: the CV RMSE lies 0.0313 below the test RMSE, about 2.4 standard errors. Part of this is expected, because λ was chosen precisely by minimising that quantity. The CV RMSE still lies inside the bootstrap interval of the test RMSE, and the CV and test values of R^2 are close. The one-standard-error rule would select $\lambda \approx 56.2$, with practically the same test performance (RMSE 0.7178, R^2 0.8804), so the exact value of λ matters little in this region.

TABLE II
RIDGE PERFORMANCE WITH $\lambda \approx 31.6$. CV VALUES ARE THE MEAN OVER 10 FOLDS $\pm$ THE STANDARD ERROR; THE TEST INTERVAL IS A 95% BOOTSTRAP INTERVAL.

Set	RMSE	R^2
Training	0.5438	0.9293
10-fold CV	0.6862 ± 0.0131	0.8822 ± 0.0077
Test	0.7175	0.8805
Test interval	[0.6457, 0.7898]	[0.8466, 0.9074]

C. Partial least squares and neural network

Fig. 3 shows the cross-validation profile of PLS in terms of RMSE and R^2. In the 10-fold CV evaluation across $A \in \{1, \ldots, 40\}$ components, the validation error drops steeply over the first ten latent factors, from an RMSE of 1.2895 ($R^2 = 0.5856$) with one component to 0.7255 ($R^2 = 0.8674$) with ten, and reaches its minimum at $A^* = 19$ components, with a mean CV RMSE of 0.7060 ± 0.0159 and R^2 of 0.8750 ± 0.0090. The profile is nearly flat between 12 and 22 components, where the differences are smaller than one standard error, and R^2 mirrors the RMSE throughout. Beyond $A = 25$, adding components slowly degrades validation performance as the projection approaches the unregularised OLS subspace. Refitted on the full training set with $A^* = 19$, PLS achieves a training RMSE of 0.5384 ($R^2 = 0.9307$) and a test RMSE of 0.7337 ($R^2 = 0.8750$, 95% bootstrap interval [0.6595, 0.8061]). Applying the one-standard-error rule selects a more parsimonious model with $A_{1\mathrm{SE}} = 11$ components, yielding a slightly lower test error (RMSE = 0.7179).

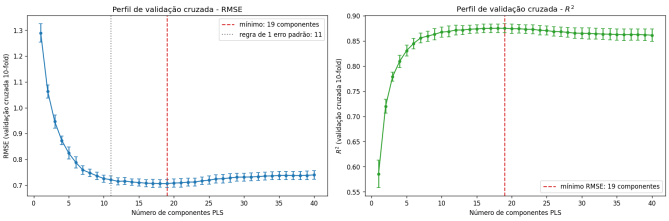


Fig. 3. PLS cross-validation profile of RMSE (left) and R^2 (right), with the standard error of the mean over 10 folds.

For the neural network, 5-fold cross-validation selected a single hidden layer of 64 units with strong L_2 weight decay (1.0), dropout rate 0.3, and 100 training epochs, achieving a mean CV RMSE of 0.6289. Strong regularisation proved essential to prevent rapid overfitting on the 951 training compounds. Fig. 4 summarises the search. With the selected weight decay and dropout, the network with 16 hidden units has the highest error at every checkpoint, and the second layer of the $(64, 32)$ architecture does not improve on a single layer. The nine best configurations lie between 0.629 and 0.637, all within one standard error (0.0204) of the best, so the error

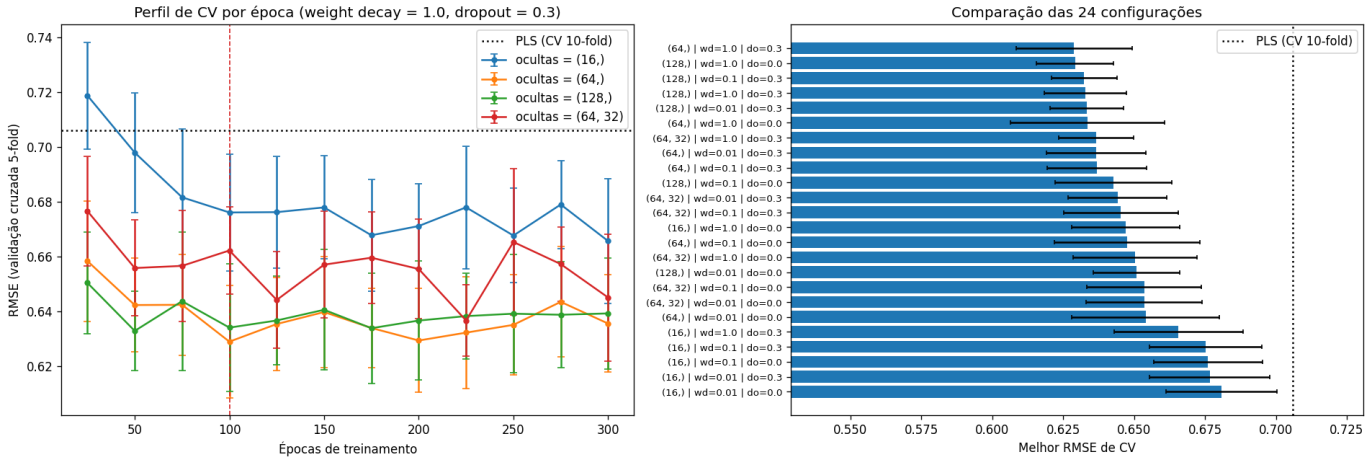


Fig. 4. Neural network 5-fold CV: RMSE over the training epochs for the four architectures with the selected weight decay and dropout (left), and best RMSE of each of the 24 configurations (right). Error bars show the standard error of the mean; the dotted line marks the 10-fold CV RMSE of PLS and the dashed line the selected 100 epochs.

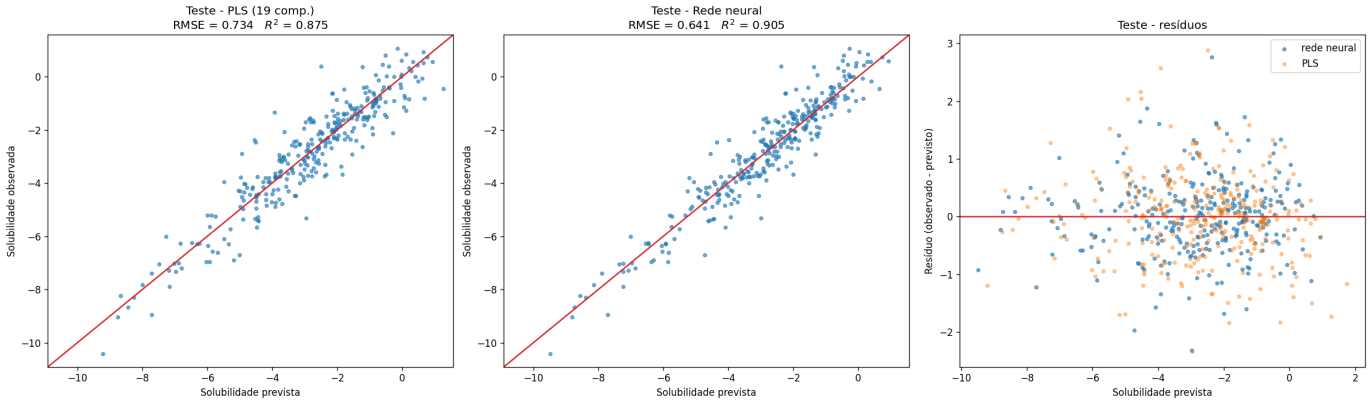


Fig. 5. Test set: observed against predicted solubility for PLS (left) and for the neural network ensemble (centre), and residuals of both models (right).

surface is flat in that region. Averaging the predictions of an ensemble of 5 independently initialised networks under this optimal configuration yielded a test RMSE of 0.6407 (95% bootstrap interval $[0.5729,\ 0.7066]$) and a test R^2 of 0.9047, substantially improving upon all linear specifications across the entire solubility range. Fig. 5 compares the test predictions with those of PLS: the observed-versus-predicted cloud of the network is tighter around the diagonal, and its residuals are smaller over most of the solubility range.

D. Statistical comparison

Table III consolidates the four models. To assess whether the differences on the test set exceed the sampling uncertainty of 316 compounds, we use a paired bootstrap [7]. Each of 5000 resamples draws the test compounds with replacement and computes the RMSE of all models on the same compounds, so that the difficulty of the resample affects every model alike. For each pair of models, Table IV gives the 95% percentile interval of $\mathrm{RMSE}_A - \mathrm{RMSE}_B$; a positive value favours model B. A difference is significant when the interval excludes zero. Since there are six simultaneous comparisons, the table also gives the interval with Bonferroni correction, at an individual level of $1 - 0.05/6$.

TABLE III
CONSOLIDATED RESULTS. CV IS 10-FOLD FOR THE LINEAR MODELS AND 5-FOLD FOR THE NEURAL NETWORK. INTERVAL: 95% BOOTSTRAP INTERVAL OF THE TEST RMSE.

Model	CV RMSE	Test RMSE	Interval	Test R^2
OLS	0.7279	0.7430	$[0.6645,\ 0.8243]$	0.8718
Ridge	0.6862	0.7175	$[0.6457,\ 0.7898]$	0.8805
PLS	0.7060	0.7337	$[0.6595,\ 0.8061]$	0.8750
Neural net.	0.6289	0.6407	$[0.5729,\ 0.7066]$	0.9047

No difference among the three linear models is significant. Ridge has the lowest test RMSE of the group, and it beat PLS in 95.8% of the resamples, but the interval of that pair, $[-0.0350,\ 0.0020]$, still contains zero, and the same holds for the pairs involving OLS. The test set therefore does not support ranking the linear models, even though cross-validation places ridge first.

The neural network is significantly better than each linear

TABLE IV
PAIRED BOOTSTRAP OF $\mathrm{RMSE}_A - \mathrm{RMSE}_B$ ON THE TEST SET (5000 RESAMPLES)

$A - B$	Diff.	95% interval	Bonferroni interval
OLS − Ridge	0.0255	[−0.0075, 0.0593]	[−0.0195, 0.0706]
OLS − PLS	0.0093	[−0.0248, 0.0455]	[−0.0366, 0.0573]
OLS − NN	0.1023	[0.0524, 0.1538]	[0.0322, 0.1715]
Ridge − PLS	−0.0162	[−0.0350, 0.0020]	[−0.0425, 0.0076]
Ridge − NN	0.0768	[0.0416, 0.1138]	[0.0268, 0.1262]
PLS − NN	0.0930	[0.0498, 0.1395]	[0.0325, 0.1564]

model, with or without the Bonferroni correction. Its test RMSE is lower by 0.1023 than that of OLS, by 0.0768 than that of ridge and by 0.0930 than that of PLS, and it won in at least 99.98% of the resamples in all three pairs. The cross-validated values point the same way, although they are not strictly comparable, since the network used 5 folds and the linear models 10.

IV. CONCLUSION

Four regression models were fitted to predict solubility from 228 molecular descriptors, with pre-processing and hyper-parameters determined on the training set only. The OLS, ridge, and PLS estimators and their cross-validation were implemented from scratch and agree with the built-in functions to numerical precision.

The three linear models reached test RMSE values of 0.7430 (OLS), 0.7175 (ridge) and 0.7337 (PLS), with R^2 between 0.8718 and 0.8805, and the paired bootstrap found no significant difference among them. Regularisation changed the coefficients far more than the predictions: ridge reduced the norm of the coefficients from 152.02 to 1.48 while lowering the test RMSE by only 0.0255.

The neural network reached a test RMSE of 0.6407 and an R^2 of 0.9047, significantly better than every linear model, with a reduction of 10.7% in RMSE against ridge, the best linear model. This indicates that the relationship between the predictors and solubility is not purely linear. Since the linear models already explain about 87% to 88% of the variance on the test set, the relationship appears predominantly linear with a smaller nonlinear component. These conclusions rest on a single test set of 316 compounds, as reflected in the width of the intervals of Table III.

USE OF AI TOOLS

AI assistance (Anthropic Claude) was used in this work for code debugging, for verification queries over the cross-validation and model results, for structuring the text and for checking the references. The prompts, the responses and the use made of each are recorded in `USO_DE_IA.md` in the project repository.

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